TABLE 6.1	
LED	**LCD**
Consumes more power (10-250 mW per digit)	Consumes less power (10-200 mW per digit)
Good brightness level	Moderate brightness level
Lifetime is around 1,00,000 hours	Lifetime is limited to 50,000 hours due to chemical degradation
Operating voltage range is from 1.5 V to 3.3 V DC	Operating voltage range is from 3 to 20 V AC
Emits light in red, green, orange, yellow, blue and white	Invisible in darkness and requires external illumination

Disadvantages of LCD

1. LCD requires an external or internal light source. (poor visibility in darkness)
2. Its life time is limited due to chemical degeneration.
3. It operates slowly. Turn-On time is few ms.

When used on DC, the life-span is quite small and therefore, they are used with AC supplies having a frequency less than 50 Hz.

Application of LCD

1. It is used in solid state video displays.
2. It is used in laptop computers.
3. It is used in pocket calculators.
4. It is used in instrument displays and digital watches.
5. It can be used as numerical counter for counting items.

Comparison of LED and LCD

Table 6.1 is the comparison between LED and LCD.

6.5. PHOTO DIODE

Silicon photo diode is a light sensitive device, also called photo detector, which converts light signal into electrical signal.

Construction: Figure 6.5 shows the construction and symbol of photo diode. The diode is made of a semiconductor PN

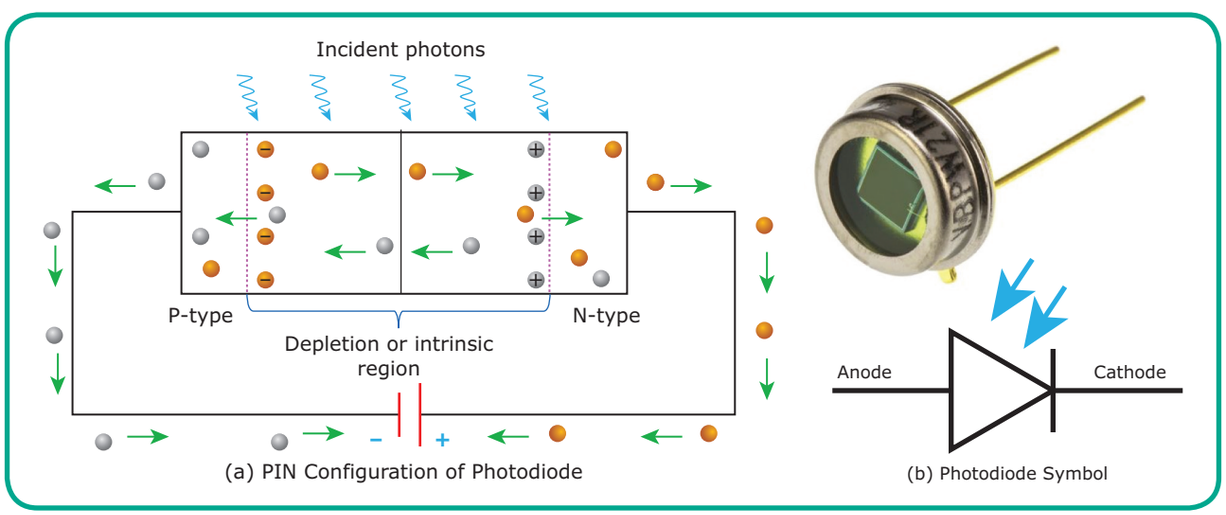


Figure 6.5 **Construction and Symbol of Photo Diode**

ELEC_EQUIP_Chapter06.indd 128 11/10/2020 11:44:08 AM